

Write a java program to implement RSA algorithm, to encrypt the data while sending it and decrypt while receiving

```
import java.math.BigInteger;
import java.util.*;
class rsa
{
    public static void main(String args[])
    {
        Scanner ip=new Scanner(System.in);
        int p,q,n,e=1,j;
        int d=1,i1;
        int t1,t2;
        int pt[]= new int[10];
        int ct[]= new int[10];
        int rt[]= new int[10];
        int temp[]= new int[10];
        String i=new String();
        System.out.println("Enter the two prime numbers:");
        p=ip.nextInt();
        q=ip.nextInt();
        System.out.println("Enter the message to be sent");
        i=ip.next();
        i1=i.length();
        n=p*q;
        t1=p-1;
        t2=q-1;
        System.out.println("\n-----");
        System.out.println("Sender Side:");
        while((t1*t2)%e==0)
        {
            e++;
        }
        System.out.println("Public Key(e)= "+e);
        System.out.println("-----");

        for(j=0;j<i1;j++)
        {
            pt[j]=(i.charAt(j))-96;
            // System.out.println("Plain Text= "+pt[j]);
            ct[j]=((int)Math.pow(pt[j],e))%n;
            System.out.println("Cipher Text= "+ct[j]);
        }
        System.out.println("\nTransmitted Message:");
        for(j=0;j<i1;j++)
        {
            temp[j]=ct[j]+96;
            System.out.print((char)temp[j]);
        }
    }
}
```

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System.out.println("\n\n-----");
System.out.println("Receiver Side:");
while((d*e)%(t1*t2)!=1)
{
    d++;
}
System.out.println("Private Key(d)= "+d);
System.out.println("-----");

for(j=0;j<i1;j++)
{
    //System.out.println("cipher Text= "+ct[j]);
    BigInteger very_big_no = BigInteger.valueOf(ct[j]);
    very_big_no = very_big_no.pow(d);
    very_big_no = very_big_no.mod(BigInteger.valueOf(n));
    rt[j] = very_big_no.intValue();
    System.out.println("Plain Text= "+rt[j]);
}
System.out.println("\n-----");
System.out.println("Decrypted Message:");
for(j=0;j<i1;j++)
{
    rt[j]=rt[j]+96;
    System.out.print((char)rt[j]);
}
System.out.println("\n-----");
ip.close();
}
}

```